Chapter 11: Solutions and Colloids

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PHY201

Lecture 11

Part I

The Dissolution Process

Outline of Part I: The Dissolution Process

What Is a Solution?

Outline of Part I: The Dissolution Process

What Is a Solution?

Solutions: A Homogeneous Mixture

Definition: Solution

A **solution** is a homogeneous mixture of two or more substances in which all components are uniformly distributed at the molecular scale.

Solvent & Solute

- **Solvent**: the component present in the greatest amount; determines the physical state of the solution.
- **Solute**: the component(s) dissolved in the solvent.
- An **aqueous solution** has water as the solvent.

Five Properties of Solutions

- Homogeneous throughout
- Physical state matches the solvent
- Dispersed at the molecular scale
- Dissolved solute does not settle
- Composition is variable (within solubility limits)

Types of Solutions

Solutions exist in all phases — not just liquid:

Solute	**Solvent**	**Example**
Gas (O_2)	Gas (N_2)	Air
Gas (CO_2)	Liquid (H_2O)	Soft drinks
Gas (H_2)	Solid (Pd)	Hydrogen in palladium
Liquid (H_2O)	Liquid (C_3H_8O)	Rubbing alcohol
Solid (NaCl)	Liquid (H_2O)	Saltwater
Solid (Zn)	Solid (Cu)	Brass

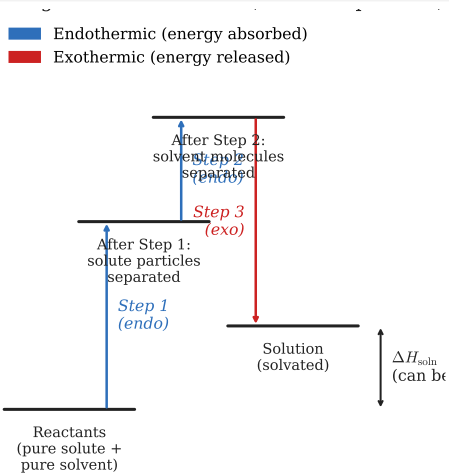
The solvent is the component present in the greatest concentration; everything else is a solute.

The Energetics of Dissolution

Dissolution can be modeled in **three steps**:

1. **(Endothermic)** Separate solute particles — break solute–solute attractions.
2. **(Endothermic)** Separate solvent molecules — break solvent–solvent attractions.
3. **(Exothermic)** Solvation — form new solute–solvent attractions.

$$\Delta H_{\text{soln}} = \Delta H_{\text{step 3}} - (\Delta H_{\text{step 1}} + \Delta H_{\text{step 2}})$$



Driving Forces for Dissolution

Two thermodynamic driving forces promote dissolution:

1. Decrease in Energy

If solute–solvent interactions release more energy than separation requires, dissolution is energetically favorable ($\Delta H_{\text{soln}} < 0$).

2. Increase in Entropy

Mixing disperses matter and increases system disorder. This thermodynamic tendency favors dissolution even when it is endothermic.

Examples

- **NaOH** in water: highly exothermic — solvation forces exceed lattice energy.
- $\mathbf{NH_4NO_3}$ in water: endothermic, yet spontaneous — driven by entropy (instant cold packs).
- $\mathbf{CaCO_3}$: lattice energy far exceeds solvation energy — essentially insoluble.

Ideal Solutions

Definition: Ideal Solution

A solution in which the solute–solvent interactions are essentially equal to the solute–solute and solvent–solvent interactions, so that $\Delta H_{\text{soln}} \approx 0$.

Examples of Ideal (or Near-Ideal) Solutions

- Helium and argon gas mixtures
- Methanol and ethanol
- Pentane and hexane

These mixtures form spontaneously due to the **increase in entropy** — even without an energy change. The intermolecular forces between the mixed species are virtually identical to those in the pure components, so mixing is essentially free energetically.

Check Your Understanding

When ammonium nitrate (NH_4NO_3) dissolves in water, the solution becomes very cold.

(a) Is this dissolution **exothermic** or **endothermic**?

(b) Does this mean it will *not* dissolve spontaneously? Explain briefly.

Come to lecture to see the answer.

Part II

Electrolytes

Outline of Part II: Electrolytes

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Electrolytes vs. Nonelectrolytes

Electrolyte

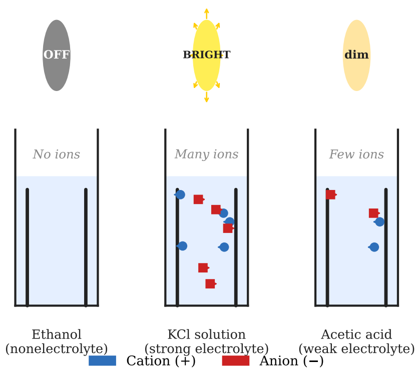
A substance that produces **ions** when dissolved in water, enabling the solution to conduct electricity.

Nonelectrolyte

A compound that dissolves without producing ions — the solution does **not** conduct electricity.

Examples: ethanol (C_2H_5OH), sucrose, glucose

Electrical conductivity is directly



Strong and Weak Electrolytes

Strong Electrolyte

Ion-producing process is essentially **100% complete**.

Examples: NaCl, KCl, HCl, HNO_3, NaOH

$$\mathrm{NaCl}(s) \xrightarrow{H_2O} \mathrm{Na}^+(aq) + \mathrm{Cl}^-(aq)$$

Weak Electrolyte

Only a **small fraction** of dissolved molecules produce ions (partial ionization).

Examples: acetic acid (CH_3COOH), NH_3

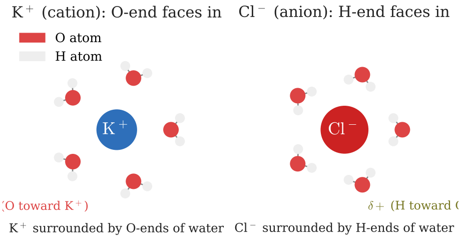
$$\mathrm{CH_3COOH}(aq) \rightleftharpoons \mathrm{H}^+(aq) + \mathrm{CH_3COO}^-(aq)$$

The double arrow ($\rightleftharpoons$) signals **equilibrium**: weak electrolytes only partially ionize. Strong electrolytes use a single forward arrow ($\rightarrow$) because ionization is

Ionic Electrolytes: Dissociation (Physical Change)

For **ionic compounds** (e.g., KCl):

1. Water molecules cluster around surface ions.
2. **Ion-dipole forces** (between the ion and polar water) overcome the electrostatic lattice forces.
3. Ions separate and disperse into solution — a **physical change**.



O-end ($\delta-$) of water orients toward **cations**;
H-end ($\delta+$) orients toward **anions**.

Covalent Electrolytes: Ionization (Chemical Change)

Covalent compounds become electrolytes by **chemically reacting** with water to produce ions — a **chemical change** (ionization).

Hydrogen Chloride (HCl) — Strong Acid

Pure HCl gas: covalent molecules, no ions, no conductivity.

Dissolved in water:

$$HCl(g) + H_2O(\ell) \longrightarrow H_3O^+(aq) + Cl^-(aq)$$

Reaction is ∼100% complete ⇒ **strong electrolyte**.

Water — Extremely Weak Electrolyte

Only ∼2 out of every **1 billion** water molecules ionize at 25 °C:

Check Your Understanding

Classify each as a **strong electrolyte**, **weak electrolyte**, or **nonelectrolyte**:

(a) NaCl (table salt)

(b) Ethanol (C_2H_5OH)

(c) Acetic acid (CH_3COOH)

(d) HCl (hydrochloric acid)

Come to lecture to see the answer.

Part III

Solubility

Outline of Part III: Solubility

Factors Affecting Solubility

Outline of Part III: Solubility

Factors Affecting Solubility

Solubility: Key Terms

Definition: Solubility

The **solubility** of a substance is its maximum concentration in solution at equilibrium, under specified conditions of temperature and pressure.

Concentration Vocabulary

- **Saturated**: at the solubility limit (equilibrium between dissolving and crystallizing).
- **Unsaturated**: below the limit — more solute can dissolve.
- **Supersaturated**: above the limit — unstable; crystallization can be triggered by mechanical shock or seeding.

Other Useful Terms

- **Dilute**: relatively low solute concentration.
- **Concentrated**: relatively high solute concentration.
- **Miscible**: liquids that mix in all proportions.
- **Immiscible**: liquids with minimal mutual solubility.

"Like Dissolves Like"

The General Rule

Substances with **similar intermolecular forces** tend to dissolve in each other:

- **Polar solvents** (e.g., water) dissolve **polar and ionic** solutes.
- **Nonpolar solvents** (e.g., hexane, CCl_4) dissolve **nonpolar** solutes.

Miscible with Water

Ethanol, sulfuric acid, ethylene glycol — all polar; can form hydrogen bonds or strong dipole interactions with water.

Immiscible with Water

Gasoline, benzene, CCl_4 — all nonpolar; weak dispersion forces cannot compete with water's strong hydrogen bonds.

Oil and water don't mix because oil is nonpolar and water's strong H-bonds with itself exclude nonpolar molecules.

Gas Solubility: Effect of Temperature

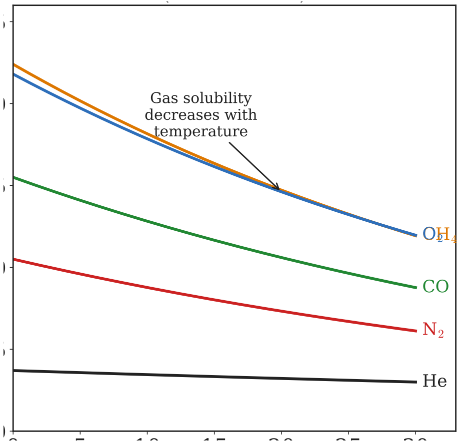
Gas solubility in liquids **decreases** as temperature increases.

As temperature rises, dissolved gas molecules gain kinetic energy to escape the solution — the equilibrium shifts toward the gas phase.

Environmental Consequence: Thermal Pollution

Warm-water discharge from power plants reduces dissolved O_2.

Freshwater trout require $\geq 7.5\,\frac{\text{mg}}{\text{L}}$ dissolved O_2 to survive — warm streams may fall



Gas Solubility: Henry's Law

Henry's Law

The concentration of a dissolved gas is **directly proportional** to its partial pressure above the solution:

$$C_g = k \cdot P_g$$

C_g = dissolved gas concentration $(\frac{\text{mol}}{\text{L}})$, P_g = partial pressure of the gas, k = Henry's Law constant (gas-, solvent-, and temperature-specific).

- **Carbonated drinks**: bottled under high CO_2 pressure; opening releases pressure $\Rightarrow$ CO_2 solubility drops $\Rightarrow$ bubbles.
- **The "bends" (decompression sickness)**: N_2 dissolves into blood under high pressure during deep dives; rapid ascent causes N_2 bubbles to form in tissues.
- **Lake Nyos (1986), Cameroon**: volcanic CO_2 had saturated the deep lake

Example: Applying Henry's Law

Henry's Law Calculation — (chem2e_ch11_ex11.2)

At 20 °C, the Henry's Law constant for O_2 in water is $k = 1.36 \times 10^{-5}\,\frac{\text{mol}}{\text{L kPa}}$.

(a) What is $[O_2]$ when $P_{O_2} = 101.3\,\text{kPa}$ (pure O_2)?

(b) What is $[O_2]$ at atmospheric conditions ($P_{O_2} = 20.7\,\text{kPa}$)?

(a) Pure O_2:

$$C_g = k \cdot P_g$$

$$= 1.36 \times 10^{-5}\,\frac{\text{mol}}{\text{L kPa}} \times 101.3\,\text{kPa}$$

$$\boxed{C_g = 1.38 \times 10^{-3}\,\frac{\text{mol}}{\text{L}}}$$

(b) Atmospheric O_2:

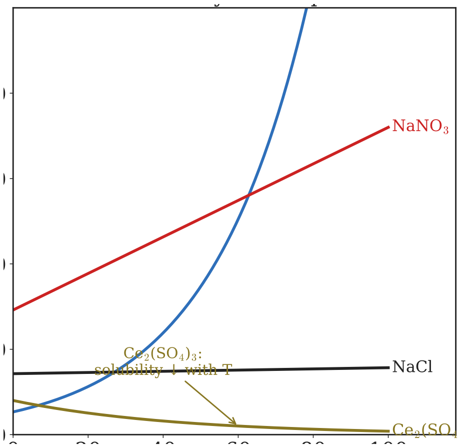
$$C_g = 1.36 \times 10^{-5}\,\frac{\text{mol}}{\text{L kPa}} \times 20.7\,\text{kPa}$$

$$\boxed{C_g = 2.82 \times 10^{-4}\,\frac{\text{mol}}{\text{L}}}$$

Solid Solubility: Effect of Temperature

For **most solids**, solubility **increases** with temperature.

- **KNO_3**: dramatically more soluble at 80 °C than 20 °C.
- **NaCl**: nearly constant over 0–100 °C.
- **$Ce_2(SO_4)_3$**: *decreases* with temperature (notable exception).



Application: Hand Warmers

Sodium acetate ($NaC_2H_3O_2$) forms a **supersaturated** solution when cooled. Clicking a metal disc triggers rapid **exothermic crystallization**, releasing

Check Your Understanding

A warm can of soda is opened and fizzes violently.

Explain this using **two** concepts from this section.

(Hint: think about both temperature and pressure.)

Come to lecture to see the answer.

Part IV

Colligative Properties

Outline of Part IV: Colligative Properties

Outline of Part IV: Colligative Properties

Colligative Properties: Overview

Definition: Colligative Properties

Properties of solutions that depend only on the **total number** of dissolved solute particles, **not** on the chemical identity of those particles.

The Four Colligative Properties

1. **Vapor pressure lowering**
2. **Boiling point elevation**
3. **Freezing point depression**
4. **Osmotic pressure**

1 mol of glucose, 1 mol of NaCl (which gives 2 mol of ions), and 1 mol of urea are chemically different — but a given *number of particles* affects colligative properties the same way, regardless of identity.

Concentration Units: Mole Fraction and Molality

Mole Fraction (X)

$$X_A = \frac{n_A}{n_{\text{total}}}$$

n_A = moles of component A
n_{total} = total moles of all components
Dimensionless; $\sum_i X_i = 1$.

Molality (m)

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}\ (\text{kg})}$$

Units: $\frac{\text{mol}}{\text{kg}}$

Temperature-independent: mass does not change with T, unlike volume (molarity).

Both mole fraction and molality are **temperature-independent** — preferred for colligative property calculations where temperature changes substantially.

Outline of Part IV: Colligative Properties

Vapor Pressure Lowering: Raoult's Law

Raoult's Law

The partial pressure of a volatile component above its solution equals the product of its **mole fraction** and its **pure-component vapor pressure**:

$$P_A = X_A \cdot P_A^*$$

For a solution with a **nonvolatile solute**:

$$P_{\text{soln}} = X_{\text{solvent}} \cdot P_{\text{solvent}}^*$$

Since $X_{\text{solvent}} < 1$, the vapor pressure above a solution is always **lower** than above the pure solvent.

Physical basis: Nonvolatile solute molecules occupy surface sites, reducing the evaporation rate of solvent.

Outline of Part IV: Colligative Properties

Boiling Point Elevation and Freezing Point Depression

Boiling Point Elevation

The lower vapor pressure of the solution means a **higher temperature** is needed to reach atmospheric pressure and boil:

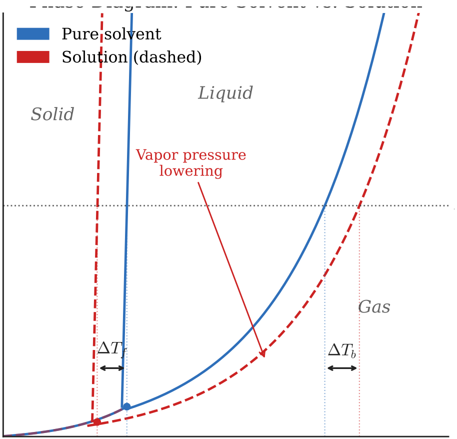
$$\Delta T_b = K_b \cdot m$$

ΔT_b = boiling point elevation (°C)
K_b = ebullioscopic constant $\left(\frac{°\mathrm{C\,kg}}{\mathrm{mol}}\right)$
m = molality $\left(\frac{\mathrm{mol}}{\mathrm{kg}}\right)$

Freezing Point Depression

A solute lowers the freezing point:

$$\Delta T_f = K_f \cdot m$$



Constants and Applications

Water Constants

$$K_b = 0.512\,\frac{^\circ\mathrm{C\,kg}}{\mathrm{mol}}$$
$$K_f = 1.86\,\frac{^\circ\mathrm{C\,kg}}{\mathrm{mol}}$$
$$\text{Normal BP} = 100\,^\circ\mathrm{C}$$
$$\text{Normal FP} = 0\,^\circ\mathrm{C}$$

Real-World Applications

- **Road de-icing**: NaCl and $CaCl_2$ lower the freezing point of water on roads.
- **Antifreeze**: Ethylene glycol in car radiators prevents both freezing and boiling.
- **Cooking**: Adding salt to pasta water raises the boiling point slightly (though minimally in practice).

Example: BP Elevation and FP Depression

Ethylene Glycol — (chem2e_ch11)

62.1 g of ethylene glycol ($C_2H_6O_2$, $M = 62.07\,\frac{\text{g}}{\text{mol}}$) is dissolved in 250 g of water. Find the boiling point elevation and freezing point depression.

Moles of solute:

$$n = \frac{62.1\,\text{g}}{62.07\,\frac{\text{g}}{\text{mol}}} = 1.00\,\text{mol}$$

Molality:

$$m = \frac{1.00\,\text{mol}}{0.250\,\text{kg}} = 4.00\,\frac{\text{mol}}{\text{kg}}$$

Boiling point elevation:

$$\boxed{\Delta T_b = 0.512 \times 4.00 = 2.05\,^\circ\text{C}}$$

Freezing point depression:

$$\boxed{\Delta T_f = 1.86 \times 4.00 = 7.44\,^\circ\text{C}}$$

New BP: $102.05\,^\circ\text{C}$

New FP: $-7.44\,^\circ\text{C}$

Outline of Part IV: Colligative Properties

Osmosis and Osmotic Pressure

Osmosis

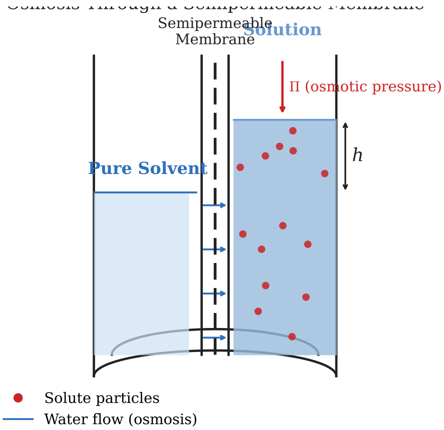
Diffusion of **solvent molecules** through a **semipermeable membrane** from lower to higher solute concentration.

Osmotic Pressure (Π)

The external pressure required to **stop** net osmosis:

$$\Pi = MRT$$

M = molarity $\left(\frac{\text{mol}}{\text{L}}\right)$
$R = 0.08206\,\frac{\text{L atm}}{\text{mol K}}$
T = absolute temperature (K)



Osmosis: Applications and Tonicity

Reverse Osmosis

Applying pressure $> \Pi$ forces solvent from the *concentrated* side to the *dilute* side:

- **Seawater desalination**
- Water purification systems

Biological Relevance: Tonicity

Blood serum $\Pi \approx 7.7\,\text{atm}$.

IV solutions must be **isotonic**:

- **Isotonic**: no net water flow; normal cell volume.
- **Hypotonic**: water flows *into* cells $\Rightarrow$ **hemolysis** (bursting).
- **Hypertonic**: water flows *out* $\Rightarrow$ **crenation** (shriveling).

Osmotic pressure is extremely sensitive to solute concentration and can be measured accurately — it is widely used to determine the molar masses of large biomolecules (proteins, polymers).

Outline of Part IV: Colligative Properties

The Van't Hoff Factor for Electrolytes

Van't Hoff Factor (i)

For electrolytes, the actual number of dissolved particles exceeds the number of formula units. The **van't Hoff factor** corrects for this:

$$i = \frac{\text{actual moles of particles in solution}}{\text{moles of formula units dissolved}}$$

Theoretical vs. Actual i

Solute	Ideal i	Actual i (0.05 m)
NaCl	2	≈ 1.9
$MgCl_2$	3	≈ 2.7
Sucrose	1	≈ 1.0

$i <$ theoretical value because of **ion pairing**: residual interionic attraction reduces the effective particle count. At higher dilution, i approaches the ideal value.

Check Your Understanding

Which solution would have the **lowest** freezing point?

(a) 1.0 m glucose ($C_6H_{12}O_6$) — nonelectrolyte

(b) 1.0 m NaCl — strong electrolyte, 2 ions

(c) 1.0 m $MgCl_2$ — strong electrolyte, 3 ions

(d) 0.5 m $MgCl_2$

Come to lecture to see the answer.

Part V

Colloids

Outline of Part V: Colloids

Colloidal Dispersions

Outline of Part V: Colloids

Colloidal Dispersions

What Are Colloids?

Definition: Colloid (Colloidal Dispersion)

A mixture in which particles are larger than typical solution molecules but small enough that they **do not settle** under gravity.

- **Dispersed phase**: the particulate component.
- **Dispersion medium**: the substance throughout which particles are dispersed.

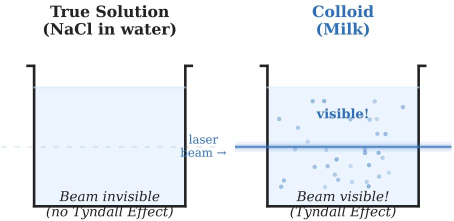
Type	**Particle Size**	**Settles?**	**Light?**
True solution	$< 1\,\mathrm{nm}$	No	Passes through (transparent)
Colloid	1–1000 nm	No	Scatters (cloudy/opalescent)
Suspension	$> 1000\,\mathrm{nm}$	Yes	Opaque

The Tyndall Effect

Colloidal particles are large enough to **scatter visible light**, making a colloid appear **cloudy or opalescent** when a beam passes through it.

Examples

- Light beam visible through fog or smoke.
- A flashlight beam is visible in milk.
- Blue color of the sky (light scattered by atmospheric particles).
- **True solutions** (saltwater, sugar water) show **no** Tyndall Effect.



Types of Colloidal Systems

Dispersed Phase	Medium	Name	Example
Solid	Gas	Smoke	Smoke, dust
Solid	Liquid	Sol	Paint, starch, milk of magnesia
Solid	Solid	—	Gems, alloys
Liquid	Gas	Aerosol	Clouds, fog, mist
Liquid	Liquid	Emulsion	Milk, mayonnaise
Liquid	Solid	Gel	Jellies, gelatin (Jell-O)
Gas	Liquid	Foam	Whipped cream, sea foam
Gas	Solid	—	Pumice

Emulsions and Emulsifying Agents

Emulsifying Agent

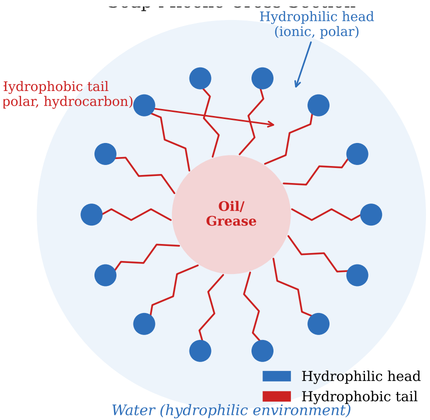
An **amphiphilic** molecule with both a **hydrophobic** (nonpolar) end and a **hydrophilic** (polar/ionic) end.

How It Works

- Nonpolar tails embed in oil droplets.
- Polar/ionic heads face the water.
- This prevents droplets from coalescing.

Examples

- **Milk**: butterfat in water; stabilized by casein protein



Soaps and Detergents

Soap

Long nonpolar hydrocarbon chain + ionic **carboxylate** group ($-COO^-$).

Example: Sodium stearate, $C_{17}H_{35}CO_2Na$

Detergent

Similar amphiphilic structure but with a **sulfate** ($-OSO_3^-$) or **sulfonate** ($-SO_3^-$) head group instead of carboxylate.

Why Detergents Beat Soaps in Hard Water

Hard water contains Ca^{2+} and Mg^{2+} ions. Soaps form **insoluble precipitates** ("soap scum") with these ions, reducing cleaning power.

Detergent sulfonate/sulfate groups form **water-soluble** products even with Ca^{2+} and Mg^{2+}.

Cleaning Mechanism

Nonpolar tails surround grease/dirt; ionic heads face water; grease is suspended as

Electrical Properties and the Cottrell Precipitator

Charge Stability of Colloids

All colloidal particles in a given system carry the **same sign of charge**:

- Metal hydroxide colloids: typically *positive*.
- Metal and metal sulfide colloids: typically *negative*.

Electrostatic repulsion between like-charged particles **prevents aggregation and settling**.

Cottrell Precipitator (Frederick Cottrell)

1. High voltage applied to electrodes in

Adding an electrolyte to a colloid can cause **coagulation**: the added ions neutralize the surface charges, removing the repulsive barrier and allowing particles to aggregate and settle.

Case Study: Deepwater Horizon (2010)

Dispersant Corexit emulsified 4.9 million barrels of oil into colloidal droplets, increasing surface area for bacterial degradation.

Check Your Understanding

A student shines a laser pointer into two cups:

(A) Saltwater (NaCl dissolved in water)

(B) A glass of milk

In which cup is the laser beam **visible from the side**, and why?

Come to lecture to see the answer.

Summary: Chapter 11 — Solutions and Colloids

- A **solution** is a homogeneous mixture; the solvent sets the phase. Dissolution is driven by energy minimization and/or entropy increase.
- **Electrolytes** produce ions (strong: complete; weak: partial); **nonelectrolytes** produce none.
- **Solubility**: "like dissolves like"; gas solubility decreases with T and follows **Henry's Law** ($C_g = kP_g$) for pressure; most solid solubilities increase with T.
- **Colligative properties** depend only on total particle concentration:
  - Vapor pressure lowering (Raoult's Law: $P_A = X_A P_A^*$)
  - BP elevation ($\Delta T_b = K_b m$); FP depression ($\Delta T_f = K_f m$)
  - Osmotic pressure ($\Pi = MRT$)
  - For electrolytes, multiply by van't Hoff factor i.
- **Colloids** (1–1000 nm particles) scatter light (Tyndall Effect), do not settle, and are used in soaps, emulsions, and industrial precipitators.